SOFTWARE REQUIREMENTS SPECIFICATION

Generative AI-Based Cloud Removal and Reconstruction
for LISS-IV Satellite Imagery

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Purpose. This document defines the expected capabilities, interfaces, constraints, and acceptance criteria for a prototype that identifies cloud-obscured areas in LISS-IV imagery and produces a model-generated reconstruction.

Important limitation: reconstructed pixels are estimates, not guaranteed recovery of ground truth. The application must label generated regions and avoid presenting them as verified observations.

Document Control and Contents

This SRS is organized in the style of a requirements specification and is intended as a practical project baseline. It is not a certification of compliance with any standard. Confirm any institution-specific template requirements before submission.

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Revision history

Version	Date	Change
1.0	October 2026	Initial SRS draft for the project concept.

1. Introduction

1.1 Purpose

This Software Requirements Specification defines what the proposed Generative AI-Based Cloud Removal and Reconstruction system for Linear Imaging Self Scanning Sensor-IV (LISS-IV) satellite imagery should do. It provides a shared reference for implementation, testing, demonstration, and evaluation. It describes required behavior without locking the team into one model architecture.

1.2 Scope

The system will allow a user to upload supported imagery, inspect available metadata, generate or provide a cloud mask, run a reconstruction model, compare original and reconstructed imagery, and export outputs. Candidate architectures include a U-Net-based generator, a conditional generative model, or a diffusion model. The final choice should be based on available data, quality, runtime, and experimental results.

1.3 Intended audience

- Frontend, backend, and machine-learning developers.
- Testers responsible for functional, performance, and security checks.
- Project supervisor and evaluators reviewing scope and evidence.

1.4 Definitions

Term	Definition
LISS-IV	Linear Imaging Self Scanning Sensor-IV satellite imaging sensor.
Cloud mask	Pixel- or region-level map identifying likely cloud-obscured pixels.
Cloud shadow	Ground area where a cloud blocks incoming sunlight.
Reconstruction / inpainting	Model-generated estimate for image content hidden or corrupted by clouds.
Inference	Running a trained model on input data to produce predictions.
Ground truth	Reference imagery or labels used for training or objective evaluation.
MAE / RMSE	Mean Absolute Error / Root Mean Squared Error.
SSIM / PSNR	Structural Similarity Index / Peak Signal-to-Noise Ratio.

1.5 Standards note

The document uses a conventional SRS structure inspired by IEEE/ISO/IEC 29148-style requirements writing. It is a project draft; verify exact formatting and compliance needs with the institution or supervisor.

2. Overall Description

2.1 Product perspective

The application consists of a user interface, backend API, image-processing pipeline, model inference component, and storage. One possible implementation is a React frontend and Java Spring Boot backend connected to a separate model inference service. A separate service is often practical when the selected deep-learning framework runs natively in Python. This is an architectural option, not a mandatory language requirement.

Component	Responsibility
Frontend	Upload imagery, show job status, visualize original/mask/output, provide downloads.
Backend/API	Validate requests, create jobs, store metadata, manage access, return result links.
Inference service	Preprocess imagery, create or use cloud masks, reconstruct obscured areas, return artifacts.
Storage	Store source imagery, masks, reconstructed outputs, metadata, and logs under defined access rules.
Evaluation module	Calculate configured quality metrics when suitable reference imagery exists.

2.2 User classes

User	Typical needs
Research user	Upload, process, compare, inspect, and download results.
Administrator	Manage configuration, model versions, storage limits, and service health.
Evaluator	Review outputs, metrics, test logs, and known limitations.

2.3 Operating environment

- Client: modern desktop browser such as Chrome, Edge, or Firefox.
- Backend: Java and Spring Boot versions compatible with the chosen runtime; verify Java 25 compatibility of all dependencies before deployment.
- Model runtime: compatible deep-learning framework and CPU/GPU environment.
- Storage: local development storage or configured server/object storage.
- Network: HTTP/HTTPS communication; deployed service-to-service requests must be restricted to authorized endpoints.

2.4 Constraints and assumptions

- Model quality depends on suitable LISS-IV imagery, cloud labels, geographic diversity, and reference images.
- Original imagery must be preserved separately from reconstructed results.
- Large images may require tiling or asynchronous processing to control memory and latency.
- Data licences, access conditions, and institutional rules must be followed.
- At project start, a trained model, dataset, or GPU is not assumed to already exist; availability must be confirmed.

3. Functional Requirements

Each requirement has a unique identifier and should be verified by test, inspection, or demonstration. Priority values: Must, Should, and Could.

ID	Requirement	Priority
FR-01	The system shall allow a user to upload a supported LISS-IV image.	Must
FR-02	The system shall validate file type/content, size, and readability before processing.	Must
FR-03	The system shall display available metadata, including dimensions, band count, format, and geospatial metadata where present.	Must
FR-04	The system shall apply documented band-order, normalization, resizing, and tiling rules required by the selected model.	Must
FR-05	The system shall generate a cloud mask or accept a validated user-provided mask.	Must
FR-06	The system should detect cloud shadows when the selected method supports them.	Should
FR-07	The system shall run a versioned reconstruction model and generate an output image.	Must
FR-08	The system shall use the mask to identify regions targeted for reconstruction.	Must
FR-09	The original uploaded image shall not be overwritten by the reconstructed output.	Must
FR-10	The system shall provide a comparison view for original image, mask, and reconstructed image.	Must
FR-11	The system should provide a mask overlay with adjustable visibility.	Should
FR-12	The system shall report job status: queued, running, completed, or failed, as applicable.	Must
FR-13	The system shall show a clear error message when upload, validation, inference, or export fails.	Must
FR-14	The system shall allow download of reconstructed imagery and masks in supported formats.	Must
FR-15	The system should export a processing summary with timestamp, model version, parameters, and output references.	Should
FR-16	When suitable reference imagery exists, the system shall calculate configured metrics such as MAE, RMSE, PSNR, and SSIM.	Must
FR-17	If no suitable reference is available, the system shall state that reference-based metrics cannot be calculated.	Must
FR-18	The system shall record model version and processing configuration for each completed job.	Must
FR-19	If accounts are enabled, users shall not access other users' private results.	Must
FR-20	The system should implement a documented retention/deletion policy for temporary files and artifacts.	Should

3.1 Main processing workflow

- User uploads imagery; the system validates it and displays available metadata.
- Backend creates a processing job and prepares the input for the model.
- Cloud detector predicts a mask, or a supplied mask is validated and aligned.
- Reconstruction model estimates content hidden by cloud-obscured regions.
- Post-processing combines estimated and valid pixels according to the documented compositing policy.
- System stores outputs and provenance, then provides visualization and downloads.
- If aligned reference data is available, evaluation metrics are calculated and labelled with the comparison region.

3.2 Required behavior rules

- Generated content must be labelled as estimated, not observed or guaranteed ground truth.
- If a mask is malformed or dimensionally incompatible, the system must stop or request correction.
- If input bands do not match model expectations, the system must reject or apply a documented conversion.
- Metric results must state whether they were computed on masked regions, the full image, or individual bands.

4. External Interface Requirements

4.1 User interface

- Provide a clear upload screen, supported-input guidance, validation feedback, and a processing action.
- Display job status and useful failure messages.
- Label original image, cloud mask, reconstructed image, and metrics distinctly.
- Provide a warning that reconstruction can introduce artefacts or plausible but incorrect details.
- Provide keyboard-accessible controls and readable labels.

4.2 Software interfaces

Interface	Required behavior
Frontend ↔ backend	REST/HTTP API; JSON metadata; multipart upload or controlled file reference.
Backend ↔ inference service	Authenticated internal request containing job ID, input reference, mask/configuration, and model selection.
Backend ↔ storage	Store/retrieve input, mask, output, metrics, and metadata with access control.
Model ↔ image libraries	Read/write documented formats while preserving band interpretation and geospatial metadata where supported.

4.3 Illustrative API endpoints

Method / endpoint	Purpose
POST /api/v1/jobs	Upload/register an image and create a processing job.
GET /api/v1/jobs/{id}	Get status, timestamps, progress if available, and error information.
GET /api/v1/jobs/{id}/results	Get artifact links, mask, output, metrics, and model version.
GET /api/v1/jobs/{id}/download/{type}	Download an allowed result artifact.
DELETE /api/v1/jobs/{id}	Delete job artifacts when policy and permissions allow.

These paths are examples. The final API contract must document request/response schemas, authorization, and status codes.

5. Data Requirements

5.1 Input data

The primary target is LISS-IV imagery. The team must define supported formats after confirming that the selected libraries preserve required image bands and metadata. GeoTIFF may be appropriate for geospatial data; PNG/JPEG may be used for simple visualization samples but should not replace multispectral/geospatial source data when those properties matter.

5.2 Validation and integrity

- Check readability, size, dimensions, band count, data type, numeric range, and available geospatial metadata.
- Reject corrupted files and unsupported band layouts with specific messages.
- Check that masks match input dimensions or can be aligned using a documented process.
- Preserve projection, transform, pixel size, and band metadata when present; otherwise label the output as visualization-only.
- Keep a source-to-output association and record the model version and preprocessing configuration.

5.3 Output artifacts

Artifact	Contents
Reconstructed image	Image containing model-estimated content in targeted regions.
Cloud mask	Raster or image mask with documented class encoding.
Optional shadow mask	Cloud-shadow mask when supported.
Processing manifest	Job ID, source identifier, time, model version, configuration, and artifact references.
Evaluation report	Metrics, reference identifier, comparison region, and caveats.
Logs	Job events and error categories without secrets or unnecessary source content.

5.4 Dataset requirements

Training and validation data should cover varied locations, seasons, cloud types, cloud fractions, land cover, and illumination. Dataset splits should avoid leakage between training and test scenes where possible. Reference images must be sufficiently aligned and temporally appropriate; landscape change can make a reference misleading.

6. Non-Functional Requirements

ID	Quality area	Requirement / verification
NFR-01	Performance	Record upload, queue, inference, and total time for documented image sizes and hardware; set target thresholds after benchmarking.
NFR-02	Scalability	Use asynchronous jobs for long-running inference where practical; unrelated requests should remain responsive.
NFR-03	Reliability	Failed jobs must not be reported as completed; partial outputs must not be labelled final.
NFR-04	Security	Validate uploads, enforce size limits, prevent path traversal, and protect private artifacts.
NFR-05	Privacy	Document retention/deletion and prevent unauthorized access to imagery and results.
NFR-06	Usability	A first-time user should be able to upload, process, and find results using clear labels.
NFR-07	Maintainability	Identify model and preprocessing versions separately and keep configuration documented.
NFR-08	Portability	Document OS, runtime, library, and model dependencies.
NFR-09	Observability	Log job transitions, duration, model version, and error category without logging secrets.
NFR-10	Transparency	Disclose that reconstructed pixels are estimates and explain metric limitations.
NFR-11	Image fidelity	Preserve valid source pixels according to the documented compositing policy.
NFR-12	Accessibility	Core workflows should support keyboard navigation and readable text contrast.

6.1 Security requirements

- Validate file content where practical; do not trust file extensions alone.
- Apply upload-size and dimension limits to reduce resource-exhaustion risk.
- Do not expose internal filesystem paths or stack traces in API errors.
- Keep credentials out of source control and logs; use HTTPS in deployed environments.
- Restrict inference endpoints to authorized internal callers.
- If authentication is not implemented, treat the prototype as a controlled demonstration and do not expose private data publicly.

7. Use Cases and Workflows

7.1 UC-01: Reconstruct a cloud-obscured image

Field	Description
Primary actor	Research user.
Preconditions	Application is available; supported image and model configuration exist.
Trigger	User uploads an image and selects Process.
Main flow	Validate input; display metadata; create job; predict/validate cloud mask; run reconstruction; store output and manifest; show comparison and downloads.
Alternative flows	Invalid input: show error. Inference failure: mark job failed. Missing reference: complete reconstruction but omit reference-based metrics.
Postconditions	Original remains unchanged; output, mask, and metadata are available under retention policy.

7.2 UC-02: Evaluate output

The user opens a completed result and optionally selects aligned reference imagery. The system validates compatibility, calculates supported metrics, and displays them with the comparison region and caveats. If the reference is incompatible, the system explains why metrics are unavailable.

7.3 UC-03: Download results

The user chooses a reconstructed image, mask, or report. The system checks that the artifact exists and that the user is authorized, then streams the file or returns a controlled link. Expired or deleted artifacts must produce a clear message.

7.4 Workflow summary

Step	Operation	Output
1	Validate source image	Validated image record
2	Preprocess / tile	Model-ready input
3	Detect or validate cloud mask	Cloud mask
4	Run generative reconstruction	Estimated obscured content
5	Composite and preserve metadata	Reconstructed image
6	Evaluate against optional reference	Metrics and report

8. Verification and Acceptance Criteria

A prototype is ready for demonstration when Must requirements are implemented or explicitly waived by the supervisor. Model-quality claims must be supported by a documented test set and must never be invented from successful application execution alone.

Test ID	Scenario	Pass criteria
AT-01	Upload supported image	Accepted; dimensions and available metadata displayed.
AT-02	Upload corrupt/unsupported file	Rejected with clear message; no inference starts.
AT-03	Run reconstruction	Job completes and exposes output, mask, and model/version metadata.
AT-04	Inference failure	Job marked failed; partial output not labelled final.
AT-05	Compare results	Original, mask, and reconstruction are labelled and viewable.
AT-06	Download output	Downloaded artifact is readable and linked to the correct job.
AT-07	Evaluate with reference	Metrics shown only if reference is valid/aligned; region and band policy documented.
AT-08	No reference supplied	System states that reference-based metrics are unavailable.
AT-09	Pixel preservation	Valid unmasked pixels remain unchanged within defined tolerance if required by policy.
AT-10	Access control	Unauthorized access to private artifacts is blocked when accounts are enabled.
AT-11	Provenance	Completed result identifies source, model version, and configuration.
AT-12	Performance	Test report records image size, hardware, and processing time.

8.1 Model evaluation protocol

- Record dataset source/licence, sensor and band configuration, and preprocessing.
- Separate training, validation, and test scenes to reduce data leakage.
- Evaluate cloud-mask precision, recall, and F1 against labelled masks when available.
- Evaluate reconstruction against suitable reference imagery using masked-region metrics and visual inspection.
- Include representative successes and failures: dense/thin clouds, shadows, and different surface types.
- Record architecture, weights/version, inference settings, and compute environment.

8.2 Suggested evaluation measures

Measure	Purpose	Caution
MAE / RMSE	Pixel error against a reference.	Sensitive to radiometric scale and normalization.
PSNR	Error-based image quality summary.	Depends on defined data range.
SSIM	Structural similarity.	Does not prove spectral or scientific validity.
Mask precision / recall / F1	Cloud-mask quality.	Needs reliable labels and defined classes.
Visual review	Inspect artefacts and implausible details.	Complements, but does not replace, quantitative evaluation.

9. Risks, Assumptions, and Constraints

Risk	Impact	Mitigation
Limited paired data	Poor generalization or plausible but incorrect generated detail.	Use suitable references, careful scene splits, and transparent limitations.
Temporal mismatch	Reference differs because the ground changed.	Prefer near-date reference imagery and document temporal gaps.
Spectral inconsistency	RGB output may not preserve multispectral value.	Define band handling and evaluate per band where applicable.
High compute demand	Long waits or memory errors.	Use asynchronous jobs, tiling, and configured size limits.
Misleading output	Generated pixels mistaken for measurements.	Label estimates, show mask, and disclose limitations.
Metadata loss	Output may not work in GIS workflows.	Preserve and test geospatial metadata.
Unreproducible claims	Evaluation cannot be repeated.	Record dataset, protocol, metrics, and model version.

9.1 Out of scope for initial prototype

- Guaranteeing exact recovery of ground features fully hidden by clouds.
- Supporting every sensor, image format, projection, and band combination.
- Replacing expert review or authoritative satellite products.
- Making operational disaster or emergency decisions without independent validation.
- Guaranteeing real-time processing for arbitrary image sizes or low-resource devices.

9.2 Decisions to resolve before implementation

- Exact LISS-IV data product, bands, and supported file formats.
- Whether cloud masks are generated automatically, supplied by users, or both.
- Selected generative architecture and evidence for that choice.
- Model deployment boundary: separate inference service or same deployment.
- Maximum upload size, retention period, benchmark hardware, and target processing time.
- Test dataset and measurable quality thresholds for acceptance.

10. Future Enhancements

- Uncertainty maps identifying regions with low model confidence.
- Additional sensors after sensor-specific validation.
- Tiled inference for large scenes with seam-aware blending.
- Batch processing and experiment/model version comparison.
- Geospatial export and richer cloud-shadow detection.
- Audit-friendly reports with visual panels, metrics, parameters, and provenance.

Appendix A. Glossary and Suggested Deliverables

Term	Definition
Cloud removal	Identification of cloud-obscured regions and estimation of underlying appearance.
Image inpainting	Filling missing or corrupted regions using context and a learned model.
Generative model	A model that synthesizes plausible content conditioned on available input.
Compositing	Combining reconstructed estimates with source pixels using a defined policy.
Georeferencing	Information linking pixels to geographic coordinates and a coordinate reference system.

Suggested deliverables

- Source code and setup instructions for frontend and backend.
- Model inference service and model/version documentation.
- Dataset and preprocessing description subject to data licence.
- Functional and non-functional test cases and results.
- Evaluation report with metric definitions and representative examples.
- Sample input, cloud mask, reconstructed output, processing manifest, and limitations statement.

End of SRS